

S6 MOCK 2 EXAMINATIONS 2011
P425/2 MATHEMATICS
PAPER 2
TIME: 3 HOURS

INSTRUCTION TO CANDIDATES:

Attempt all the **eight** questions in section **A** and not more than **five** from section **B**.

Show all the necessary working up to the last step of the solution.

Mathematical tables and squared papers are provided.

Silent Simple non-programmable calculators may be used.

State the degree of accuracy of questions attempted using calculator or tables.

Indicate **Cal** for calculator or **Tab** for Mathematical tables.

SECTION A: (40 marks)

1. If two events A and B are independent and $3P(A \cup B) = 5P(B) = 4P(A)$
Find
(i) $P(A)$
(ii) $P(A^1 \cap B^1)$
2. A mass of 2kg is attached to a string of length 20cm. The string is fixed at the other end and it is made to rotate at 10 rads^{-1} to form a conical pendulum
Find the
(a) tension in the string,
(b) angle between the string and the downward vertical.
3. Given that $x = 15.2$, $y = 0.45$, $Z = 8.6$. Find the percentage error in $\frac{xy}{Z}$
4. Of the 30 drivers interviewed, 9 have been involved in a car accident at sometime. Of those who have been involved in the accident, 5 wear glasses. The probability of wearing glasses, given that the driver has not had a car accident is $\frac{1}{3}$.
Find the;
(a) probability that a person chosen at random wears glasses,
(b) probability that a glasses' wearer has been a crash victim.
5. A body with initial velocity $(2\mathbf{i} - 7\mathbf{j}) \text{ ms}^{-1}$ is given an impulse causing its velocity to change to $(-8\mathbf{i} + \mathbf{j}) \text{ ms}^{-1}$. If the mass of the body is 0.5 kg and the duration of the impulse is 0.2 seconds. Calculate the magnitude of the constant force involved.

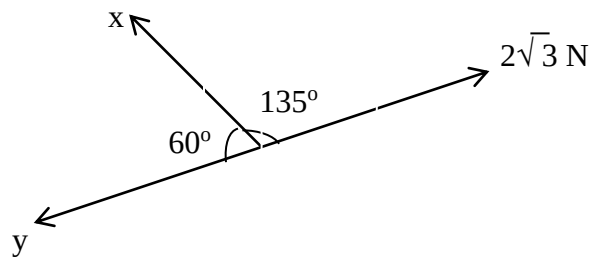
6. Show that the equation $e^{4-x} = x$ has only one real root between 2.5 and 3. Hence use linear interpolation once to find the root of the equation. Give your answer correct to three significant figures.

7. The table below shows the time (in seconds) taken to solve a certain problem

Time(s)	30 – 49	50 – 64	65 – 69	70 – 74	75 – 99
No of students	10	30	25	20	15

Draw a histogram for the data and use it to estimate the mode.

8. The figure below shows three forces in equilibrium



Find the values of x and y

SECTION B: (60 MARKS)

9. A random variable X has the cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{4}, & 0 \leq x \leq 1 \\ ax + b, & 1 \leq x \leq 2 \\ \frac{1}{4}(5-x)(x-1), & 2 \leq x \leq 3 \\ 1 & 3 < x \end{cases}$$

(a) Find

- the values of the constants a and b
- $P(0.5 \leq x \leq 2.5 / x > 1)$
- The p.d.f, $f(x)$

(b) Sketch the graph of $f(x)$ and hence deduce the mean of x .

10. (a) Given that x and y are values measured with possible errors, E_1 and E_2 respectively. Show that the maximum possible relative error in the expression $y\sqrt{x}$ is $\frac{1}{2} \left| \frac{E_1}{x} \right| + \left| \frac{E_2}{y} \right|$

Given that $x = 0.35$ and $y = 7.6$. Find the percentage error in evaluating $7.6 \sqrt{0.35}$

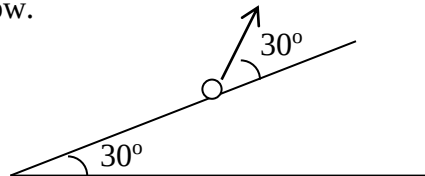
- (b) Find the maximum and minimum values in evaluating $5^{5.01}$

11. A bus of mass 18×10^3 kg travels up a slope of $\sin^{-1}(1/50)$ against a resistance of 0.1 N kg^{-1} . Find the tractive force required to produce an acceleration of 0.05 ms^{-2} and the power which is developed when the speed is 10 ms^{-1} .
A second bus of mass 25×10^3 kg experiencing the same resistance and with a maximum power 120 kw follows the first bus up the slope. If the first bus maintains the same power while on the slope, will the distance between the buses decrease when both are travelling at maximum speed?

12. The heights of students at a certain university are normally distributed. 1% of the students are over 200 cm tall and 76% are between 165cm and 200 cm tall. Find the;
(a) mean and variance of the distribution,
(b) percentage of the students who are under 158cm tall.

13. By plotting the curves $y = e^{2x}$ and $y = 5 - 4x$ on the same axes, show that the root of the equation $e^{2x} = 5 - 4x = 0$ lies between 0 and 1. Use Newton Raphson method to determine the root correct to 2 significant figures.

14. A small parcel of mass 2 kg moves on a rough plane inclined at an angle 30° to the horizontal as shown below.



The parcel is pulled up a line of greatest slope of the plane by means of a light rope which is attached to it. The rope makes an angle of 30° with the plane. If the coefficient of friction between the parcel and the plane is 0.4, and the tension in the rope is 24N.

- (a) Find the acceleration of the parcel.
If the rope now breaks, and the parcel slows down and comes to rest,
(b) (i) show that when the parcel comes to this position of rest, it immediately starts to move down the plane again.
(ii) find the acceleration of the parcel as it moves down the plane after it has come to this position of instantaneous rest.

15. The lengths, in cm, of some species of trees are known to be normally distributed with variance 1.33 cm. If a sample of 40 trees is found to have a mean of 10.2cm.
- Find the 80% confidence limits for the mean length of the population of trees.
 - Find the minimum size of the sample of trees which must be taken if the width of the symmetrical 98% confidence interval for the population mean is at most 1.5cm.
16. (a) Three particles of masses $3m$, $5m$ and λm are placed at points with coordinates $(4,0)$, $(0, -3)$ and $(4,2)$ respectively. The centre of mass of the system of the three particles is at $(2,k)$. Find the values of λ and k .
- (b) A square lamina ABCD of side $2l$ is made of a uniform thin material, when a semi-circular piece with AB as diameter is removed from the square, show that the centre of gravity of the remainder of the lamina is at a distance $\frac{20l}{3(8-\pi)}$ from AB
- If the remainder of the lamina is suspended from a light string attached at C and hangs in equilibrium. Find the angle that CD makes with horizontal.

END